

ADI Internship

mmWave CMOS Gain-Control Circuits: Survey and Design Report

Part I surveys two published CMOS VGAs; Part II and Part III present my own VGA (built on a previously-designed LNA), and DSA designs.

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ADI Internship — VGA & DSA Assignment

Part I — Literature Survey

A comparison of two FR3-band CMOS VGA designs relevant to 5G mmWave.

1. Introduction

A variable-gain amplifier (VGA) is a core building block in 5G receiver front-ends. It compensates for changing signal strength, sets receiver dynamic range, and — in phased arrays — tunes element gain independently of phase to shape the array's sidelobes. A good VGA needs a wide, well-controlled gain range, low power consumption, low noise figure (NF), and, in phased-array applications, minimal insertion-phase change as the gain is varied ("phase invariance").

This survey compares two CMOS VGAs designed for the K/Ka-band (millimeter-wave) region adjoining 5G FR2 and the proposed FR3 mid-band extension:

- [Chang & Lin, "7.4-mW, 22.7–29.7 GHz CMOS VGA with 23.7 dB Gain Tuning Range and 3.26 dB NFavg," IEEE MWCL, 2022 — optimized for low power and low noise.](#)
- [Wu et al., "A 20–43 GHz VGA with 21.5 dB Gain Tuning Range and Low Phase Variation for 5G Communications in 65-nm CMOS," IEEE RFIC, 2019 — optimized for phase invariance across gain states.](#)

Each paper trades off a different axis of VGA performance, which makes them a useful pair for understanding the underlying circuit and layout mechanisms that govern gain and phase behavior in mmWave CMOS VGAs.

2. Chang & Lin (2022) — 7.4-mW, 22.7–29.7 GHz CMOS VGA

Published in IEEE Microwave and Wireless Components Letters, this design targets 28-GHz 5G NR band N257 (26.5–29.5 GHz) and prioritizes low power and low noise while still achieving a wide gain range.

2.1 Circuit topology

The VGA is a three-stage cascaded common-source (CS) amplifier fabricated in 90-nm CMOS (Fig. 1a). Stage 1 (M1) is a fixed-gain low-noise input stage; Stages 2 and 3 (M2, M3, with auxiliary transistor M4) form the current-reused, variable-gain back end.

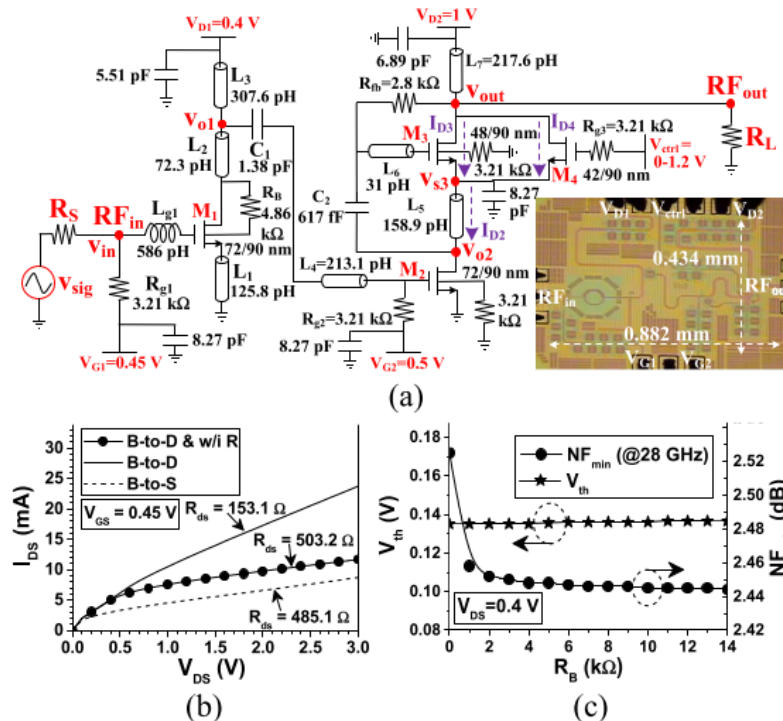


Fig. 1 (Chang & Lin, 2022): (a) VGA schematic and chip photograph; (b) I_{DS} - V_{DS} curves showing the effect of body-floating resistance R_B ; (c) V_{th} and NF_{min} vs. R_B for transistor M1.

2.2 Core techniques

- Self-body-to-drain bias (B-to-D with R): connecting transistor M1's body to its own drain through a large floating resistor RB (4.86 kΩ) lets the body float at RF while sitting at a forward bias ($V_{BS} \approx V_{DS} = 0.4$ V) at DC. This lowers the threshold voltage (V_{th} drops to 0.136 V, 0.062 V below the B-to-S case) and suppresses substrate leakage/noise, raising R_{ds} from 153.1 Ω (plain B-to-D) to 503.2 Ω and cutting NFmin to 2.45 dB at 28 GHz — 0.08 dB lower than without RB. Because V_{DS} is low (0.4 V), the parasitic body-source diode stays off, so no current flows through RB and this benefit costs no extra power.
- Current-reused, three-stage CS topology: Stage 1 draws 6 mA from a 0.4-V rail; Stages 2/3 draw 5 mA from a 1-V rail, reusing bias current between stages rather than stacking a fresh current branch for each — the whole chip runs on just 7.4 mW.
- Auxiliary current steering (ACS): transistor M4, placed in parallel with output-stage transistor M3, steers bias current away from the signal path as V_{ctrl} rises from 0 to 1.2 V. Because gain reduction happens by starving the output transistor of current rather than by adding lossy attenuator elements, the tuning range can be very large (up to 23.7 dB) without extra noise or power penalty.
- Reactive peaking for bandwidth: parallel peaking (L3/Ceff1, L5/Ceff2) boosts the 23–25 GHz midband, while series peaking (L4, Cgs2) extends gain up to the 27–29 GHz upper corner, giving 7 GHz of 3-dB bandwidth from a low-order stage count.

2.3 Small-signal gain and noise model

The paper derives closed-form expressions for gain and noise. The forward gain is modeled as three cascaded second-order transfer functions (one per stage), each contributing an L-C resonance:

$S_{21} = 2 \cdot \left(\frac{v_{o1}}{v_{sig}}\right) \cdot \left(\frac{v_{o2}}{v_{o1}}\right) \cdot \left(\frac{v_{out}}{v_{o2}}\right)$ — a product of three second-order band-pass terms, each set by an L-C tank (ω_0 , ω_{01} , ω_{02}) and its quality factor (Q, Q1, Q2).

The noise factor F is modeled as a sum of a broadband resistive-loss term $\frac{R_g + R_{lg} + R_{ss} + R_{l1}}{R_S}$, a gate-induced-noise term proportional to C_{gs1}^2 and inversely proportional to $gm1RS$, and a second-order frequency-shaped term that peaks near the input-matching resonance. The authors deliberately set the matching-network Q slightly above the critically-damped value 0.707 (to 1.11) so the NF-vs-frequency response is slightly under-damped — this flattens NF across the band rather than letting it peak sharply at one frequency.

2.4 Measured results

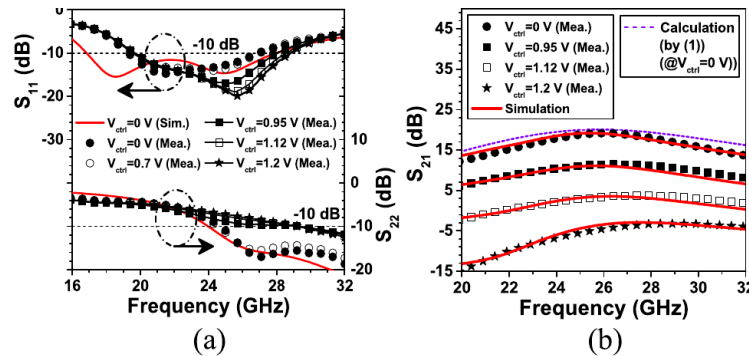


Fig. 2: (a) Measured/simulated S_{11} and S_{22} ; (b) measured, simulated, and calculated S_{21} across V_{ctrl} settings.

At $V_{ctrl} = 0$ V (max gain), $S_{21,max}$ reaches 19.2 dB at 26 GHz with 7 GHz of 3-dB bandwidth (22.7–29.7 GHz). As V_{ctrl} sweeps 0→1.2 V, the second/third-stage gain collapses from +13.6 dB to −10.1 dB at 26 GHz, while the first stage's 5.6 dB contribution stays fixed — giving an overall S_{21} tuning range of 23.7 dB at 26 GHz (19.2 to −4.5 dB), 21.3 dB at 28 GHz, and 25.2 dB at 22 GHz. S_{11} stays below −10 dB across roughly 19–29 GHz depending on V_{ctrl} , and S_{22} stays below −10 dB from 24.7 GHz upward at max gain.

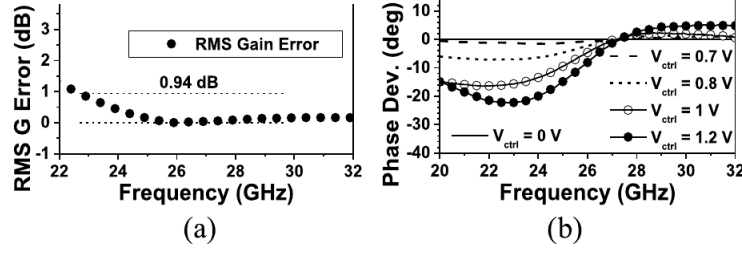


Fig. 3: Measured (a) RMS gain error for 4-bit, 1.6 dB/step gain control; (b) phase deviation (PD) across gain settings.

With 4-bit digital gain control (1.6 dB/step), the RMS gain error stays below 0.94 dB across the full band and below 0.15 dB over the N257 band of interest — i.e., the ACS gain steps are highly linear and repeatable. Phase deviation stays within $\pm 4.5^\circ$ over N257, worsening to $-22.3^\circ \dots +4.5^\circ$ only at the lowest-gain setting ($V_{ctrl} = 1.2$ V), where most of the signal current has been steered away from the output stage and the residual output impedance/parasitic path dominates the phase response.

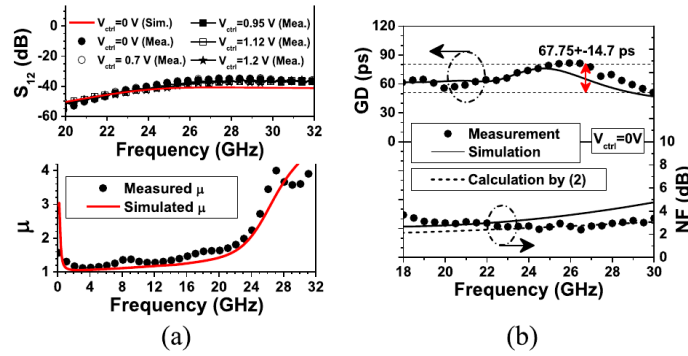


Fig. 4: (a) Measured/simulated S_{12} and stability factor μ ; (b) group delay (GD) and NF at $V_{ctrl} = 0$ V.

S_{12} stays below -34.5 dB across 20–30 GHz and the stability factor μ exceeds 1.08 across 0–32 GHz, so the VGA is unconditionally stable over a very wide band — important since a VGA with marginal stability could oscillate at some gain setting even if it behaves at others. Group delay varies by only ± 14.7 ps over 18–30 GHz (± 7.5 ps simulated), attributed to the L3/L4/L5 peaking inductors pushing S_{21} 's poles outside the band of interest. NFmin of 2.39 dB and NFavg of 2.76 dB are achieved over 22.5–29.5 GHz, close to the calculated values (2.43 dB / 2.72 dB).

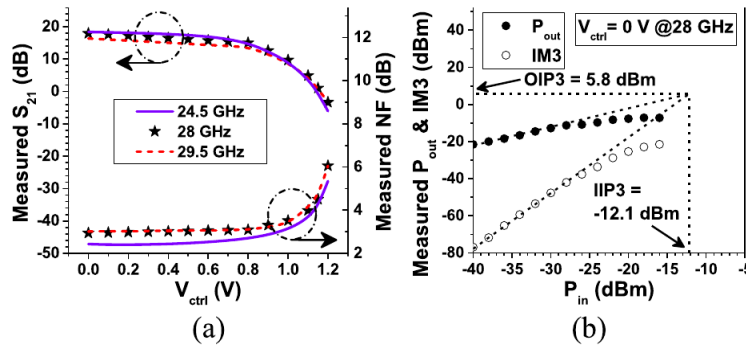


Fig. 5: (a) Measured S_{21} and NF vs. V_{ctrl} at 24.5/28/29.5 GHz; (b) measured output power and IM3, giving OIP3.

Across the full V_{ctrl} range (0–1.2 V), NF stays at 2.4–3 dB at max gain and rises only to 3.26–3.46 dB (NFavg) as gain is tuned down — and even in the "off" state ($V_{ctrl} = 1.2$ V, $S_{21} = -3.4$ to -6.1 dB) NF is still a respectable 5.34–6.05 dB, better than competing designs. The measured OIP3 is a moderate 5.8 dBm at 28 GHz; the authors note that OIP3 could be pushed to 12.5 dBm by raising the drain voltages (VD_1 to 0.8 V, VD_2 to 1.1 V), at the cost of higher PD (14.8 mW instead of 7.4 mW) — illustrating the direct power/linearity trade in this topology.

SUMMARY OF THE VGA AND RECENTLY REPORTED STATE-OF-THE-ART VGAs WITH SIMILAR OPERATION FREQUENCY

	Freq. (GHz)	S ₁₁ (dB)	S ₂₁ (dB)	f _{3dB} (GHz)	Gain Tuning Range (dB)	NF _{min} /NF _{avg} (dB)	OIP3 (dBm)	P _D (mW)	FOM (GHz/mW)	Process
This Work	28	< -10	17.7±1.5	7	23.7 (-4.5~ 19.2)	2.39/ 3.26	5.8	7.4	6.49	90 nm CMOS
[3], 2019 MWCL	18.7	< -10	19.7±1.5	3.5	5 (16.2~21.2)	NA	NA	8	NA	65 nm CMOS
[4], 2020 TMTT	31.4	< -9.5	9±1.5	25.3	16 (-5.5~ 10.5)	2.4/ 7.2	9.6*	25	0.67	45 nm CMOS
[5], 2020 IMS	33.5	< -10	11±1.5	11	18 (-5.5~ 12.5)	NA	14.1*	45	NA	120 nm SiGe
[6], 2020 TCAS-II	24.5	< -10	27±1.5	5	6 (22.5~28.5)	3.3/ 4.2	-2.5	48	1.43	130 nm SiGe
[7], 2020 IMS	24	< -5	28.4±1.5	3.4	35.4 (-5.5~ 29.9)	5.63/ 8.75	-0.5*	25.6	0.54	28 nm CMOS
	33	< -10	30.9±1.5	3.1	37.9 (-5.5~ 32.4)	4.55/ 7.46	0*		0.93	
	50	< -10	20.7±1.5	14.8	33.7 (-11.5~ 22.2)	5.96/ 10.5	-2.2*		0.61	

*Estimation: OIP3=IIP3+Gain (or IP_{1dB}+9.6+Gain)

Table I: Chang & Lin's VGA vs. contemporary state-of-the-art VGAs at similar frequencies.

The authors' figure of merit, $FOM = \frac{S_{21}[1] \cdot BW[GHz]}{(NF-1)[1] \cdot PD[mW]}$, rewards designs with high gain-bandwidth product per unit of excess noise and power. This VGA reports the lowest NF and the highest FOM (6.49 GHz/mW) among the compared works, at the cost of a narrower operating band than some competitors.

3. Wu et al. (2019) — 20–43 GHz VGA with Low Phase Variation

Presented at the IEEE RFIC Symposium, this 65-nm CMOS design targets phased-array systems, where gain must be tunable without disturbing the signal's insertion phase, so beam pointing stays fixed as sidelobes are tapered.

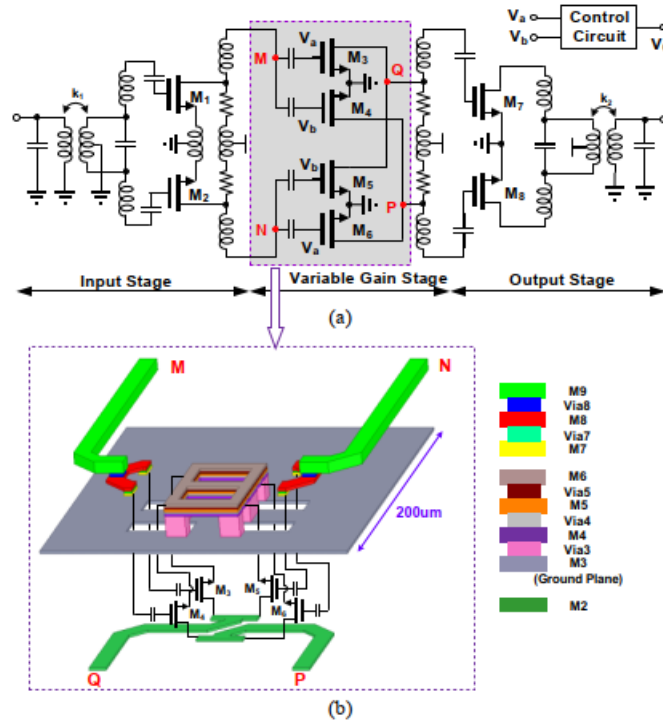


Fig. 1 (Wu et al., 2019): (a) The schematic of the proposed VGA based on parasitic capacitor elimination technique; (b) The isolation-enhanced layout of the variable gain stage in VGA.

3.1 Why phase varies in a conventional CS VGA

The paper starts from a single CS amplifier stage (Fig. 2) with gate-drain parasitic capacitance C_{gd} , drain-source capacitance C_{ds} , and output impedance r_o . Its voltage transfer function is:

$$\frac{V_{out}}{V_{in}} = \frac{gm_9 - j\omega C_{gd9}}{-\left(\frac{1}{Z_L} + \frac{1}{r_{o9}}\right) + j\omega C_{gd9} + j\omega C_{ds9}}$$

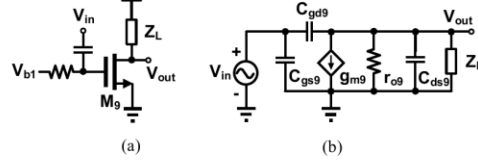


Fig. 2. Schematic of a CS amplifier and its small signal equivalent circuit.

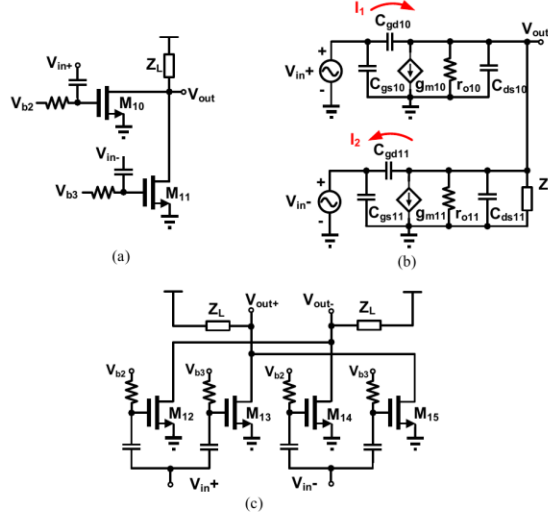


Fig. 3. Simplified schematic of the proposed variable gain amplifier (a) and its small signal equivalent circuit (b); (c) The full schematic of the proposed

Fig. 2–3 (Wu et al., 2019): (top) single-ended CS amplifier and small-signal model; (bottom) the proposed differential variable-gain stage that cancels the C_{gd} feedforward path.

When this stage is used as a VGA, the gain ratio between two bias/gain states is $\Delta G = (g_{m9,1} - j\omega C_{gd9}) / (g_{m9,2} - j\omega C_{gd9})$. Because C_{gd9} is fixed while g_{m9} changes with the gain setting, ΔG is a complex number whose phase angle depends on frequency — this is the root cause of phase variation in a conventional single-ended CS VGA.

3.2 Parasitic-capacitor elimination technique

The proposed variable-gain stage (Fig. 3) uses two identically-sized transistors (e.g., M_{10} , M_{11}) driven by a differential pair of input signals (V_{in+} , V_{in-}), so that the feedforward currents through their matched C_{gd10} and C_{gd11} are equal in magnitude but opposite in phase. These currents cancel at the shared output node, and the resulting transfer function becomes:

$$\frac{V_{out}}{V_{in+} - V_{in-}} = \frac{g_{m10} - g_{m11}}{\frac{1}{Z_L} + \frac{2}{r_{o10,11}} + 2j\omega C_{gd10,11} + 2j\omega C_{ds10,11}}$$

The gain-variation ratio is then $\Delta G = (g_{m10,1} - g_{m11,1}) / (g_{m10,2} - g_{m11,2})$ — a purely real quantity, independent of ω . In principle this makes the VGA's gain change with no phase rotation at all, and — because the denominator of ΔG can in theory reach zero — the achievable gain-tuning range is unbounded by this mechanism alone.

3.3 Isolation-enhanced layout

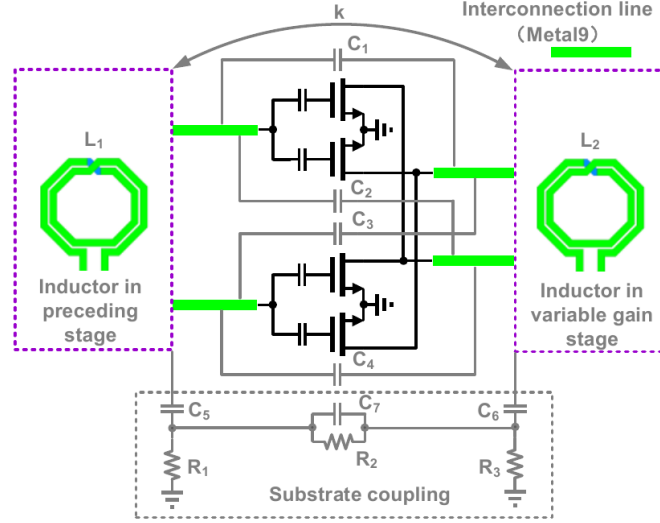


Fig. 4: Parasitic capacitors (C1–C4) formed between interconnect lines, and inductive/substrate coupling (k, R1–R3) between adjacent stages.

Transistor-level cancellation alone isn't sufficient: interconnect lines at the stage's input/output nodes create additional parasitic capacitances (C1–C4), and neighboring inductors couple magnetically (k) and through the lossy substrate (C5–C7, R1–R3). All of these are feedforward paths with high-pass characteristics, so their effect is small at low frequency but grows as frequency increases — exactly the trend that shows up in the measured phase-variation data at the high end of the band. To suppress them, the layout (Fig. 1b) routes the variable-gain stage's input nodes (M, N) on Metal9/Metal8 and its output nodes (P, Q) on Metal2, separated by a grounded Metal3 shield plane, and spaces the preceding-stage and variable-gain-stage inductors 200 μm apart.

3.4 Gain control circuit

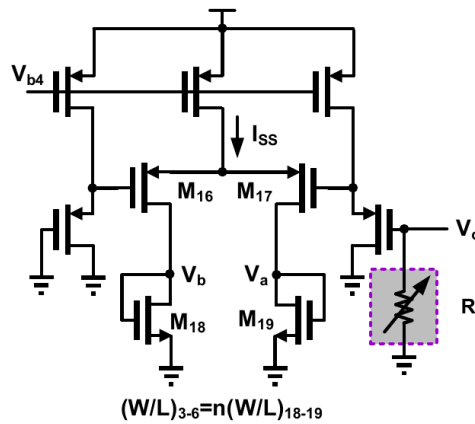


Fig. 5: Control-voltage generator producing the differential bias voltages V_a , V_b from a single external control voltage V_c .

The bias voltages V_a and V_b that steer current between the differential transistor pairs are generated from a single external control voltage V_c via a current-mirror-based generator (M16–M19), sized so that $(W/L)_{3-6} = n \cdot (W/L)_{18-19}$. The paper shows the variable-gain stage's transfer function is directly proportional to V_c , giving a simple, monotonic, single-knob gain control.

3.5 Measured results

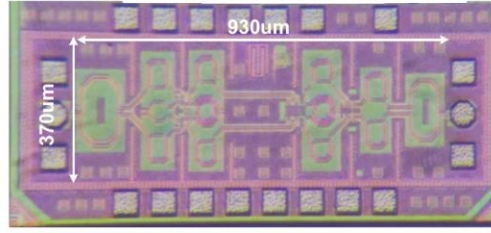


Fig. 6. Chip photo of the proposed VGA.

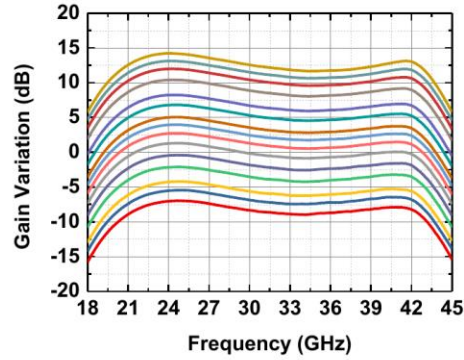


Fig. 7. Measured gain curves for various gain settings.

Fig. 6–7: Chip photograph ($370\ \mu\text{m} \times 930\ \mu\text{m}$ core) and measured gain curves for various digital gain settings across 18–45 GHz.

The chip occupies a $370\ \mu\text{m} \times 930\ \mu\text{m}$ core in 65-nm CMOS. Measured gain variation is 21.5 dB across 18–45 GHz, with a 3-dB bandwidth of 20–43 GHz and only 2.5 dB of gain ripple within that band — peak gain is 14.5 dB at 23.5 GHz.

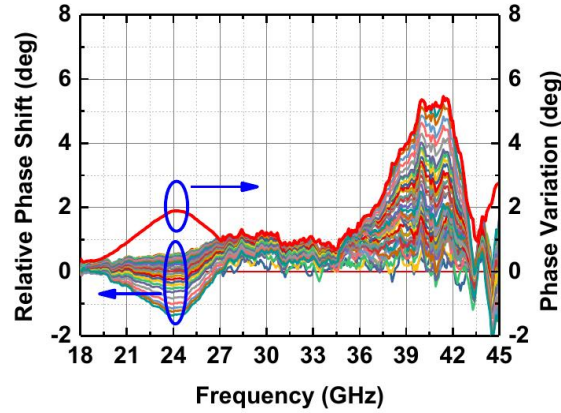


Fig. 8. Measured relative phase shift and maximum phase variation under

Fig. 8: Measured relative phase shift (left axis) and maximum phase variation (right axis) vs. frequency across all gain settings.

This is the paper's headline result: phase variation stays within just 0.2° – 2° across 18–37 GHz while gain is swept over the full 21.5 dB range, confirming that the differential Cgd-cancellation technique works as modeled. Above 37 GHz, phase variation rises, peaking at 5.4° around 39–42 GHz — consistent with the high-pass nature of the residual layout-parasitic coupling paths described in Section 3.3, which the isolation-enhanced layout suppresses but cannot eliminate entirely at the top of the band.

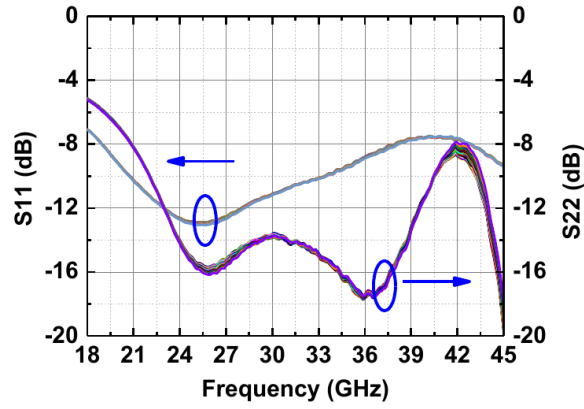


Fig.9. Measured input and output port return loss for various gain settings.

Fig. 9: Measured input (S11) and output (S22) return loss for various gain settings.

Both input and output return loss stay below -7 dB across the 3-dB bandwidth for all gain states, indicating that impedance matching is reasonably preserved as the differential pairs are steered between gain extremes — though not as deep as -10 dB, hinting that some matching compromise was accepted to prioritize phase flatness and bandwidth.

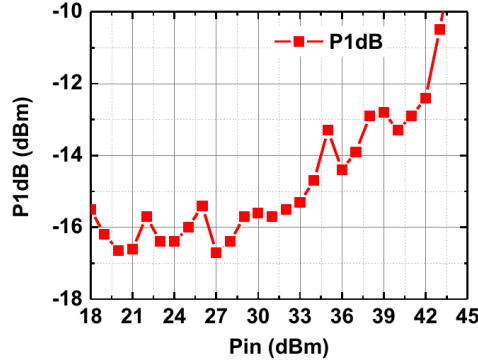


Fig. 10. Measured input-referred P1dB as the VGA in the maximum gain state.

Fig. 10: Measured input-referred P1dB in the maximum-gain state.

Input-referred P1dB is -16.5 to -10.5 dBm across the 3-dB bandwidth; the paper attributes the spread to gain unflatness across the different stages of the VGA rather than to a single dominant nonlinearity source.

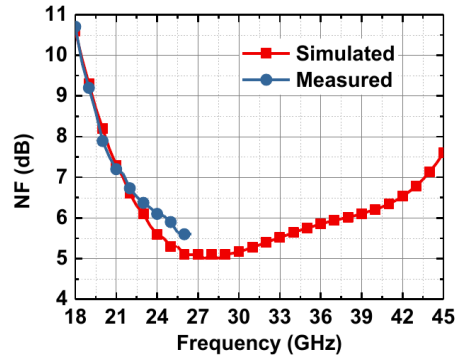


Fig. 11. Measured and simulated noise figure when VGA is at the maximum

Fig. 11: Measured vs. simulated noise figure in the maximum-gain state.

Measured NF is 5.5 – 8.2 dB across the 3-dB bandwidth, consistent with simulation. Because the design's transistors and topology are chosen primarily to enable differential Cgd cancellation and layout isolation — not for minimum-noise matching as in the Chang & Lin design — NF is noticeably higher than a noise-optimized single-ended LNA-style front end.

Table 1. Performance comparison summary

	This Work	JSSC2016[4]	RFIC2016[5]	RFIC2018[6]	ESSCIRC2016[7]
Technology	65nm CMOS	SiGe	SiGe	65nm CMOS	SiGe
Frequency	20–43GHz	10~14.4GHz	28GHz	27~42GHz	15.5~30GHz
Peak Gain	14.5dB	13dB	22dB	9.6dB	17dB
ΔG	21.5dB	22dB	18dB	7.5dB	23dB
Phase Variation	0.6°~2° #	1.2°~2°	<5°	0.7°~3.5° *	1.5°~3°
NF	5.5~8.2dB *	5.1~6.8dB	-	-	3.6~9dB
P_{dc}	30.8mW	83mW	35mW	15.6mW	104mW
Chip Area	0.34mm ²	0.7mm ²	-	0.08mm ²	0.33mm ²

< 37GHz *RMS phase error * 20 GHz ~ 26.5GHz

Table 1: Wu et al.'s VGA vs. contemporary SiGe and CMOS phase-invariant VGAs.

Against other phase-invariant VGAs (mostly in SiGe technology), this design reports the widest 3-dB bandwidth (20–43 GHz) and one of the lowest phase-variation figures (0.6°–2°, restricted to <37 GHz), while being the first to achieve this using CMOS rather than SiGe BJTs — the paper notes that the SiGe techniques in prior work rely on BJT-specific local feedback or pole-zero compensation that don't translate directly to MOSFETs.

4. Analysis: How the Gain and Phase Variation Behave

4.1 Chang & Lin — gain vs. control voltage and frequency

The VGA's small-signal gain is modeled as a cascade of three second-order resonant stages (Eq. 1 in the paper), each contributing a peaking term from an L-C tank. Physically, only the 2nd/3rd stages are tunable: at $V_{ctrl} = 0$ V, transistor M4 is off and all bias current flows through the output transistor M3, giving maximum gain (13.6 dB at 26 GHz from stages 2+3 combined). As V_{ctrl} rises to 1.2 V, M4 turns on and steers current away from M3 — the gain of those stages collapses to about –10 dB at 26 GHz. The first stage's gain (5.6 dB at 26 GHz) never changes with V_{ctrl} , since it has no ACS device, so the tuning is confined to the back two stages. Because ACS is a current-domain steering effect rather than a change in transistor bias impedance, the tuning range is very wide (23.7 dB) but its shape is frequency-dependent: it is widest at 22 GHz (25.2 dB) and narrowest at 28 GHz (21.3 dB) because the L-C peaking networks (L3/Ceff1 in midband, L4/Cgs2 near the upper corner) do not scale uniformly with V_{ctrl} .

4.2 Chang & Lin — noise vs. resistor value

Fig. 1(c) of the paper shows NFmin dropping sharply as the body-floating resistor RB increases from 0 to about 4 k Ω , then flattening — RB above a few k Ω is enough to make the body float at RF while keeping VBS fixed at DC, so extra resistance buys little further noise benefit. This is why the design settles on RB = 4.86 k Ω rather than pushing it higher — going further would add area and leakage risk for negligible NF improvement.

4.3 Wu et al. — where the phase variation comes from

Eq. (2) of the paper shows that for a plain CS-stage VGA, $\Delta G = (gm1 - j\omega Cgd)/(gm2 - j\omega Cgd)$. Because Cgd is fixed and gm changes with the gain setting, this ratio is a complex number whose amplitude and phase both vary with frequency — this is what rotates the phase as gain is tuned. The differential topology (Eq. 3–4) cancels the $j\omega Cgd$ terms between the matched transistor pair, leaving ΔG purely real and, in principle, frequency-independent. In practice this cancellation is only as good as the layout symmetry, so residual coupling — not the transistor Cgd — becomes the dominant error term at high frequency.

4.4 Wu et al. — why phase variation still grows with frequency

The measured phase variation (Fig. 8) stays under about 2° up to 37 GHz but rises to a peak near 5.4° around 39–42 GHz. The paper attributes this to the layout-parasitic paths (interconnect capacitive coupling C1–C4, inductor-to-inductor magnetic coupling k, and substrate coupling through R1–R3) that were not part of the ideal transistor-level cancellation. All of these parasitic paths behave as high-pass feedforward elements, so their effect is small at low frequency and grows as frequency increases — which is exactly the upward trend seen in the data, and is also why the paper's isolation-enhanced layout (separate metal layers, grounded shield, 200 μ m inductor spacing) is necessary rather than optional.

4.5 Net trend comparison

In the Chang & Lin design, the quantity being controlled and reported is gain magnitude — it varies smoothly and by design over more than 23 dB, with no attempt to hold phase constant, so its phase/group-delay ripple (± 14.7 ps) is treated only as a secondary linearity metric. In the Wu et al. design, gain and phase are decoupled by construction: gain still swings over 21.5 dB, but phase is actively held flat by circuit symmetry at low-to-mid frequency and only escapes that flatness where layout parasitics take over at the high end of the band. In short, one paper optimizes the amount of gain variation available; the other optimizes what does not vary (phase) while gain is varied.

5. Side-by-Side Comparison

Parameter	Chang & Lin (2022)	Wu et al. (2019)
Process	90 nm CMOS	65 nm CMOS
Frequency range	22.7 – 29.7 GHz	20 – 43 GHz
Peak gain (S21,max)	19.2 dB	14.5 dB
Gain tuning range	23.7 dB (19.2 to -4.5 dB)	21.5 dB
Noise figure	NFmin 2.39 dB, NFavg 2.76 dB	5.5 – 8.2 dB
Phase variation	Not a design target (GD var. ± 14.7 ps)	0.2° – 2° (18–37 GHz); 5.4° max across 3-dB BW
Input P1dB / IIP3	IIP3 = -12.7 dBm @ 28 GHz, OIP3 = 5.8 dBm	P1dB = -16.5 to -10.5 dBm (3-dB BW)
Power consumption	7.4 mW	30.8 mW
Chip area	0.383 mm ²	0.34 mm ²
Key innovation	Self-body-to-drain (B-to-D with R) bias + auxiliary current steering (ACS)	Parasitic-capacitor elimination + isolation-enhanced layout for phase invariance
Primary design goal	Low power, low noise, wide gain range	Low phase variation across gain states

The two designs illustrate a classic trade-off. Chang & Lin optimize for power and noise, using body-floating bias and current reuse to reach 7.4 mW and sub-3-dB average NF, at the cost of a narrower band and no explicit phase-invariance control. Wu et al. optimize for phase stability over a very wide band, using differential cancellation and careful layout isolation to hold phase variation under a few degrees across 20–43 GHz — an essential property for phased-array beamforming — but at roughly 4 $\times$ the power and a higher NF. Neither design directly targets the 3GPP FR3 band itself (7.125–24.25 GHz); both operate in the K/Ka mmWave range used by 5G FR2 (n257/n258/n261) and above, but the underlying circuit techniques — body-floating bias for noise/power efficiency, and differential parasitic cancellation for phase invariance — are broadly applicable to any CMOS VGA design problem, including future FR3 front-ends that will likely need both good NF and gain-phase orthogonality for beamforming.

6. Survey Conclusion

Both VGAs demonstrate that mmWave gain control in CMOS ultimately comes down to managing parasitics: Chang & Lin exploit body-terminal biasing to shape transistor behavior for gain/noise/power efficiency, while Wu et al. exploit differential symmetry and layout isolation to neutralize the parasitic feedforward path responsible for phase variation. The choice between them depends on system priorities — low-power/low-noise single-channel receivers favor the Chang & Lin approach, while phase-coherent phased-array elements favor the Wu et al. approach.

Part II — VGA Design

Assignment

Starting from a previously-designed LNA (10.7–12.7 GHz), design a 4 dB variable-gain amplifier by adding a switched attenuator array, and report $S_{21}/S_{11}/S_{22}$, NF, OIP3, and OP1dB across frequency and gain states, plus DNL, INL, and phase error across gain states.

7.1 Schematic and design variables

The VGA is realized by adding a 5-state switched attenuator array (0–4 dB, 1 dB/step) to the existing LNA core. The overall schematic, key transistor/matching-element design variables, and the switching (attenuator) array itself are shown below.

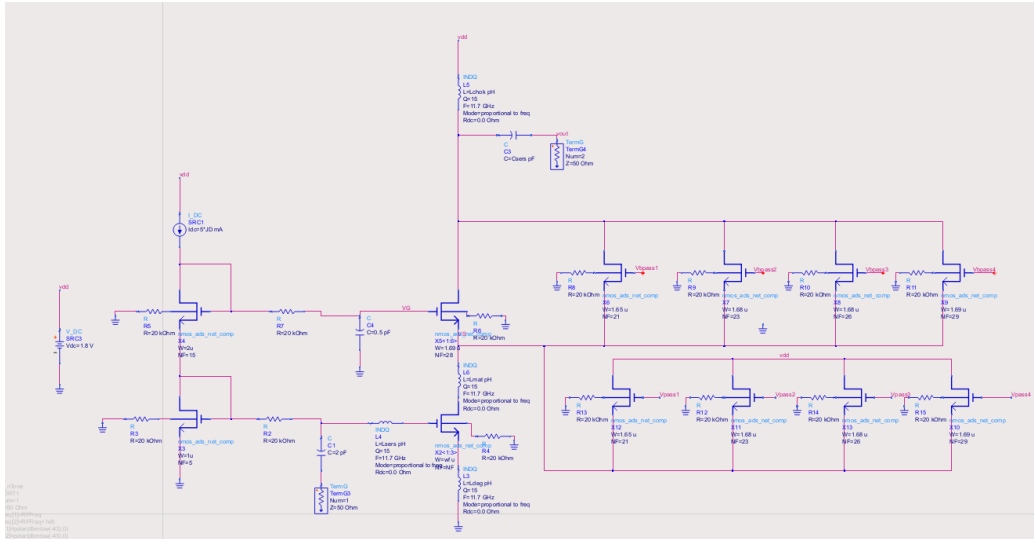


Fig. II-1: Full VGA schematic — LNA core (10.7–12.7 GHz) with the switched attenuator array added at the gain-control node.

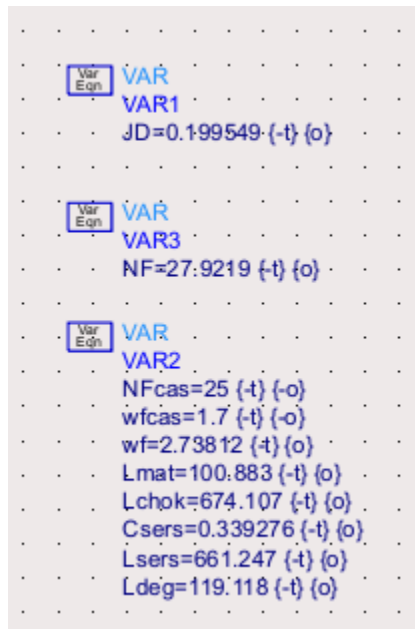


Fig. II-2: Key design variables (transistor widths/fingers, matching inductors $L_{mat}/L_{chok}/L_{sers}/L_{deg}$, series cap C_{sers}) used in the schematic.

7.2 S-parameters and noise figure vs. frequency and gain state

All S-parameter and NF sweeps below cover the design band 10.0–13.5 GHz for all 5 switch states (SW = 1...5, i.e., 0–4 dB attenuation).

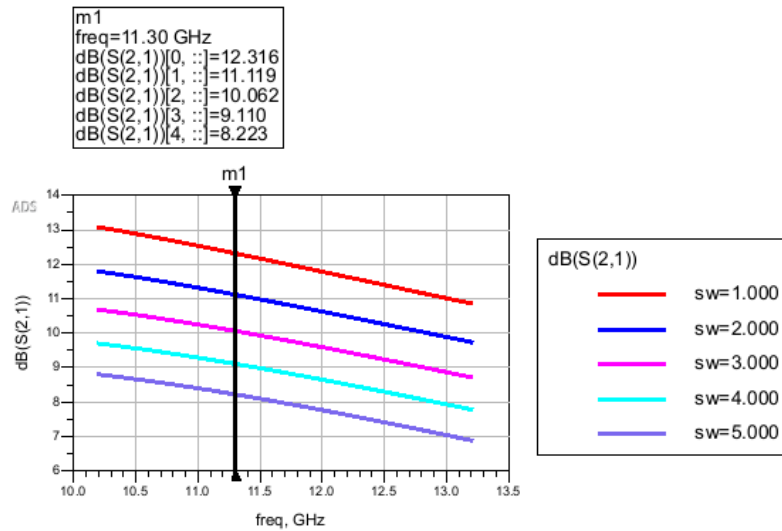


Fig. II-5: S21 (gain) vs. frequency, 10.0–13.5 GHz, for SW = 1–5 (0–4 dB attenuation).

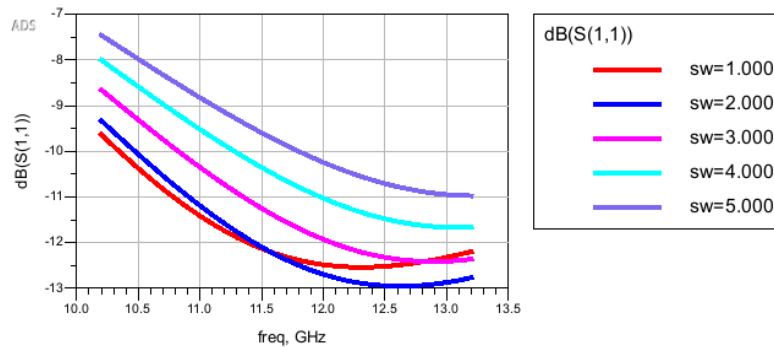


Fig. II-6: S11 (input return loss) vs. frequency, 10.0–13.5 GHz, for SW = 1–5.

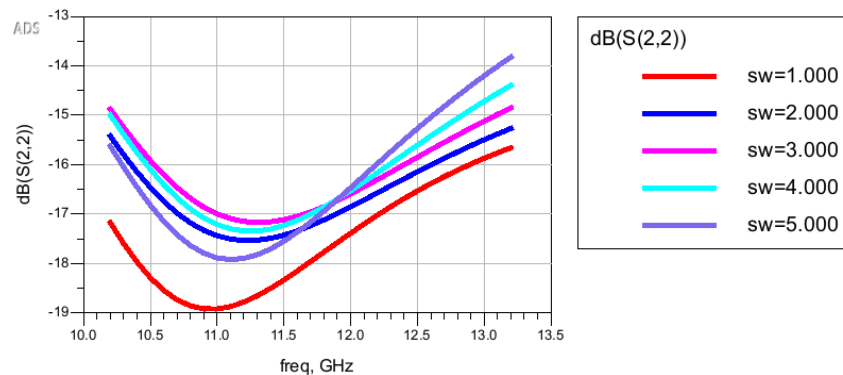


Fig. II-7: S22 (output return loss) vs. frequency, 10.0–13.5 GHz, for SW = 1–5.

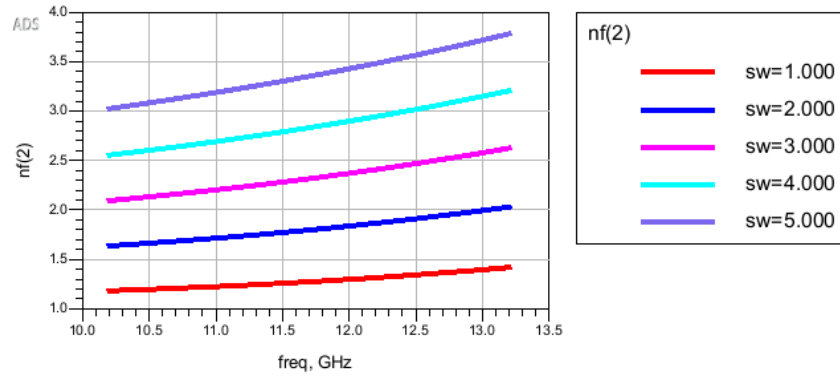


Fig. II-8: Noise figure (NF) vs. frequency, 10.0–13.5 GHz, for SW = 1–5.

7.3 Gain linearity, DNL, INL, and phase error at 11.3 GHz

The following four plots are evaluated at 11.3 GHz (near band center) across the 5 switch states, to characterize how closely the realized gain steps track an ideal 1-dB/step ladder.

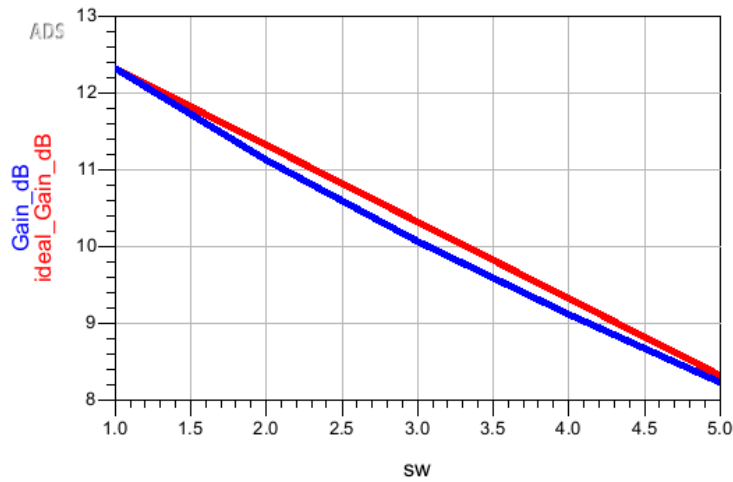


Fig. II-9: Measured gain & ideal (linear 1-dB/step) gain vs. switch state, at 11.3 GHz.

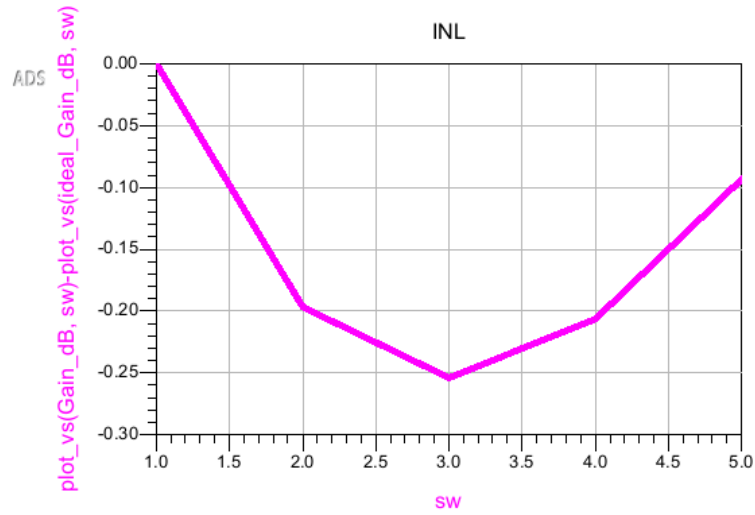


Fig. II-10: INL (measured gain – ideal gain) vs. switch state, at 11.3 GHz.

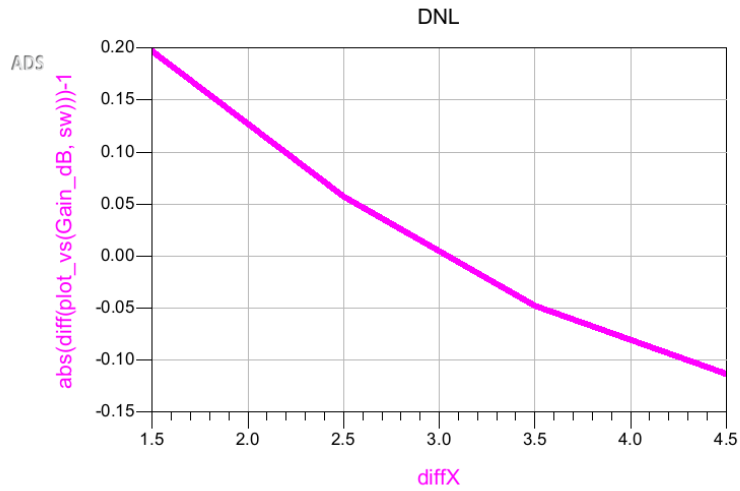


Fig. II-11: DNL (deviation of each 1-dB step from exactly 1 dB) vs. switch-state transition, at 11.3 GHz.

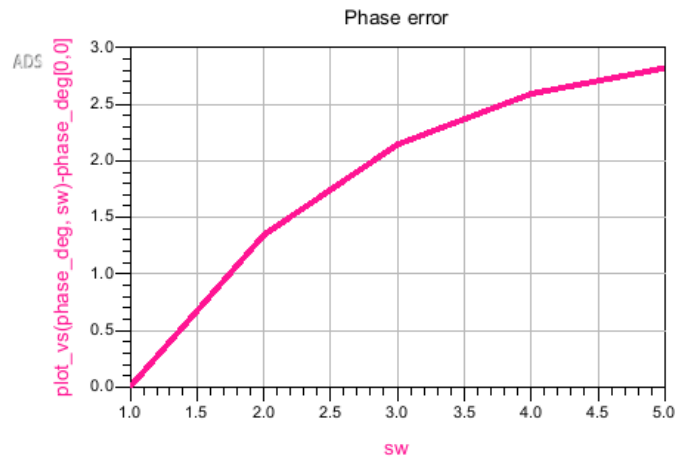


Fig. II-12: Phase error (relative to SW = 1) vs. switch state, at 11.3 GHz.

7.4 Linearity vs. frequency and gain state

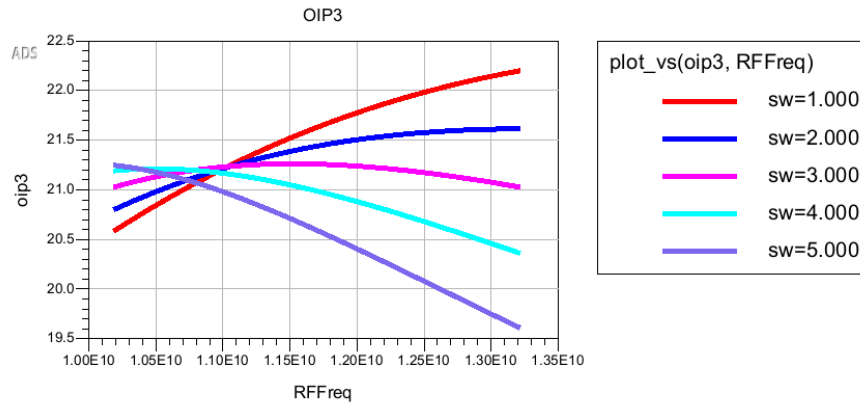


Fig. II-13: OIP3 vs. RF frequency, 10.0–13.5 GHz, for SW = 1–5.

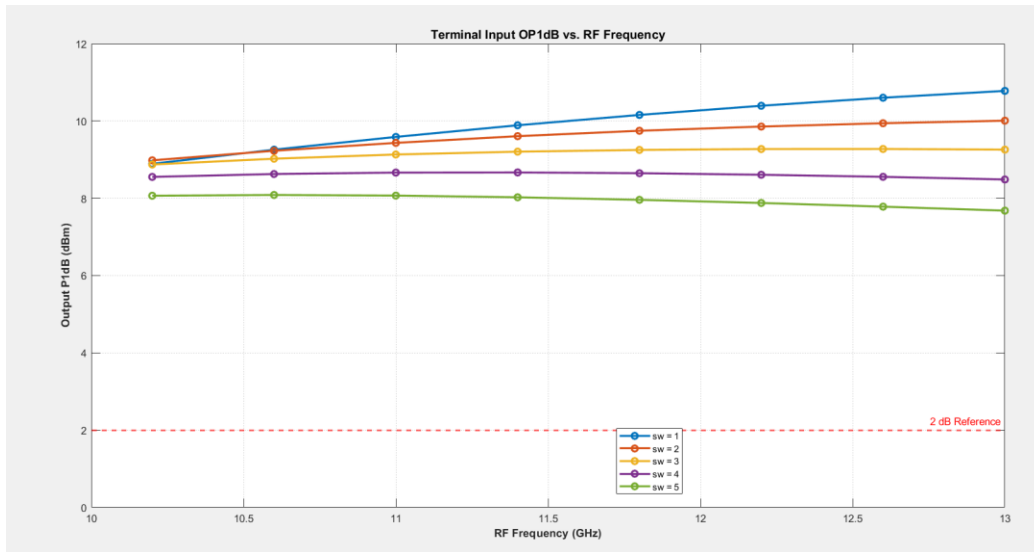


Fig. II-14: Output-referred P1dB (OP1dB) vs. RF frequency, 10.0–13.5 GHz, across switch states.

7.5 VGA summary

The design meets the 4-dB gain-tuning-range target using a 5-state (1-dB step) switched attenuator added to the existing LNA, with S11/S22 well matched and NF/OIP3/OP1dB degrading gracefully as attenuation increases — the expected behavior for a passive-attenuator-based VGA, where noise and linearity trade directly against attenuation state rather than against a separate active gain-control device.

Part III — DSA Design

Assignment

Design a digital step attenuator (DSA) to the following specification, using a cascade of 1-dB, 2-dB, and 4-dB switched attenuator sections (3-bit, 8 states, 0–7 dB range in 1-dB steps), and report schematic, testbench, and simulations: S21 vs. frequency across attenuation states; NF vs. attenuation state @ 28 GHz; DNL vs. attenuation state @ 28 GHz; gain/attenuation range vs. frequency; phase error vs. attenuation state @ 28 GHz; OP1dB vs. frequency at the minimum-attenuation state; and OP1dB vs. attenuation state @ 28 GHz.

Spec.	Value	Unit
Frequency	24 – 30	GHz
IL (min. attenuation)	2.5	dB
OIP3	—	dBm
Gain range	7	dB
Gain step	1	dB
Return loss	–10	dB
Supply	1.2	V
Interface	Single-ended	—

8.1 Schematic and design variables

The DSA cascades three binary-weighted switched-attenuator sections (nominally 1 dB, 2 dB, 4 dB) so that an 8-level control word ($sw = 0 \dots 7$) produces attenuation from 0 to 7 dB in 1-dB steps — matching the 7-dB gain-range / 1-dB gain-step specification.

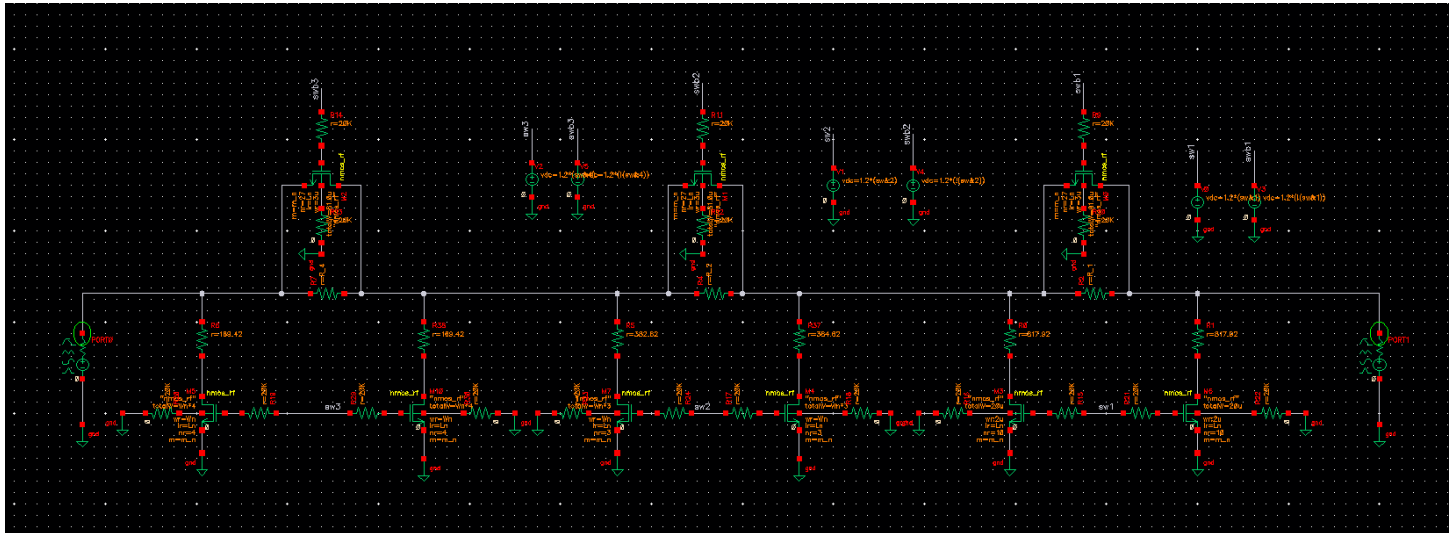


Fig. III-1: DSA schematic — cascaded 1-dB/2-dB/4-dB switched attenuator sections, single-ended, 1.2-V control.

Design Variables	
L_n	60n
m_n	1
nf_n	10
sw	0:1:7
W_n	2u
R_1	17
R_2	18.16
R_4	32

Fig. III-2: Design variables — switch transistor length/width/fingers (L_n , W_n , nf_n), attenuator resistor values (R_1 , R_2 , R_4), and the 8-level sweep $sw = 0:1:7$.

8.2 S21 (gain/attenuation) vs. frequency across states

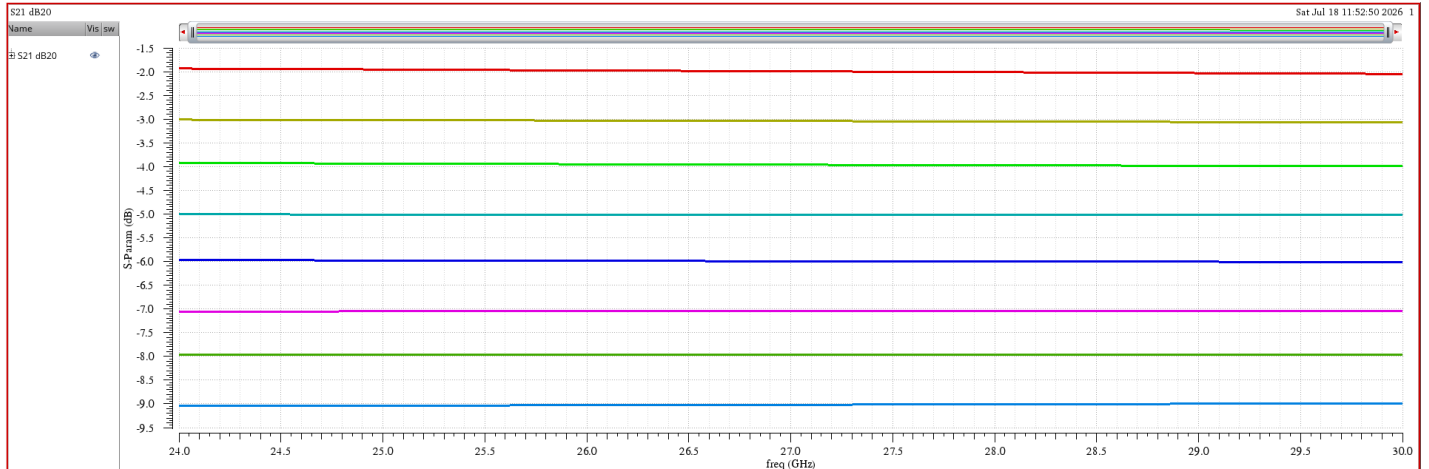


Fig. III-3a: S21 vs. frequency, 24–30 GHz, for all 8 attenuation states ($sw = 0-7$).

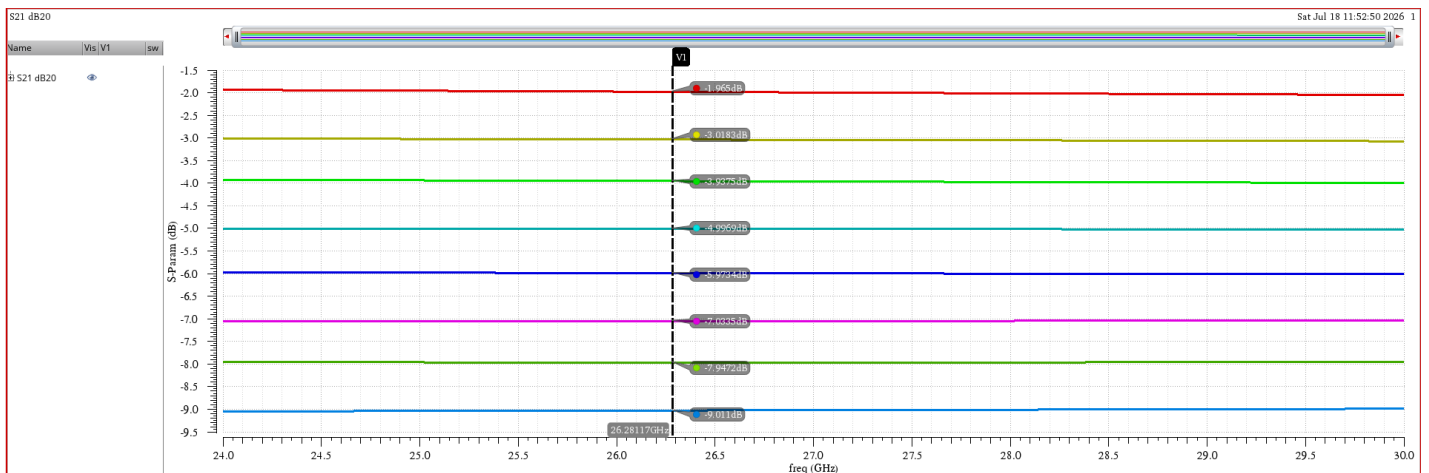


Fig. III-3b: S21 vs. frequency, 24–30 GHz (marker readout: -1.965 dB (insertion loss), $sw = 7$).

8.3 Noise figure and dynamic range vs. frequency

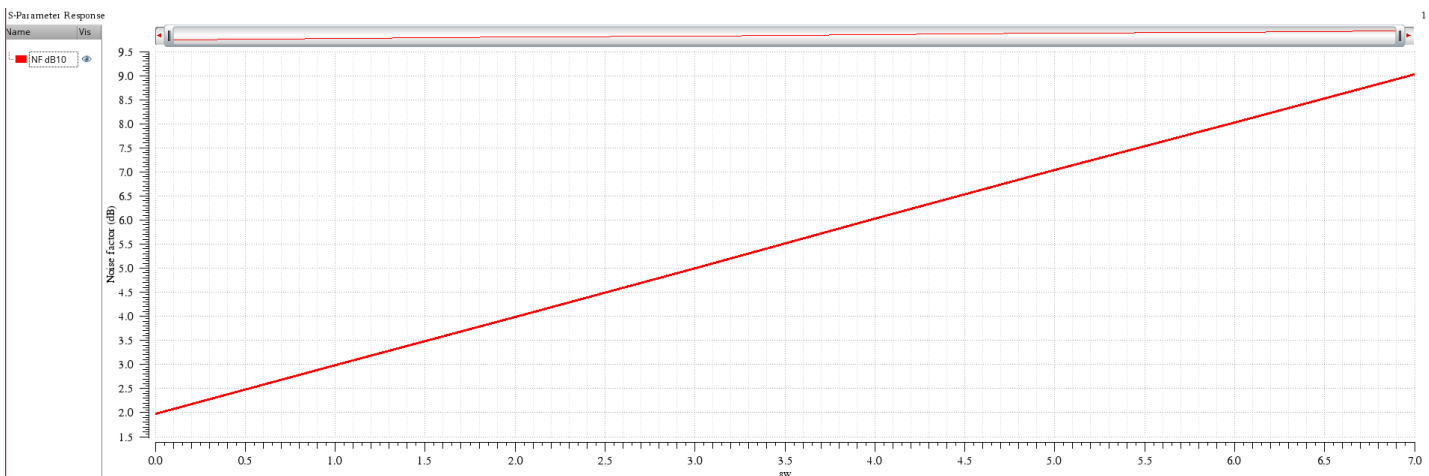


Fig. III-4: Noise figure vs. frequency, 24–30 GHz, across attenuation states ($sw = 0-7$).

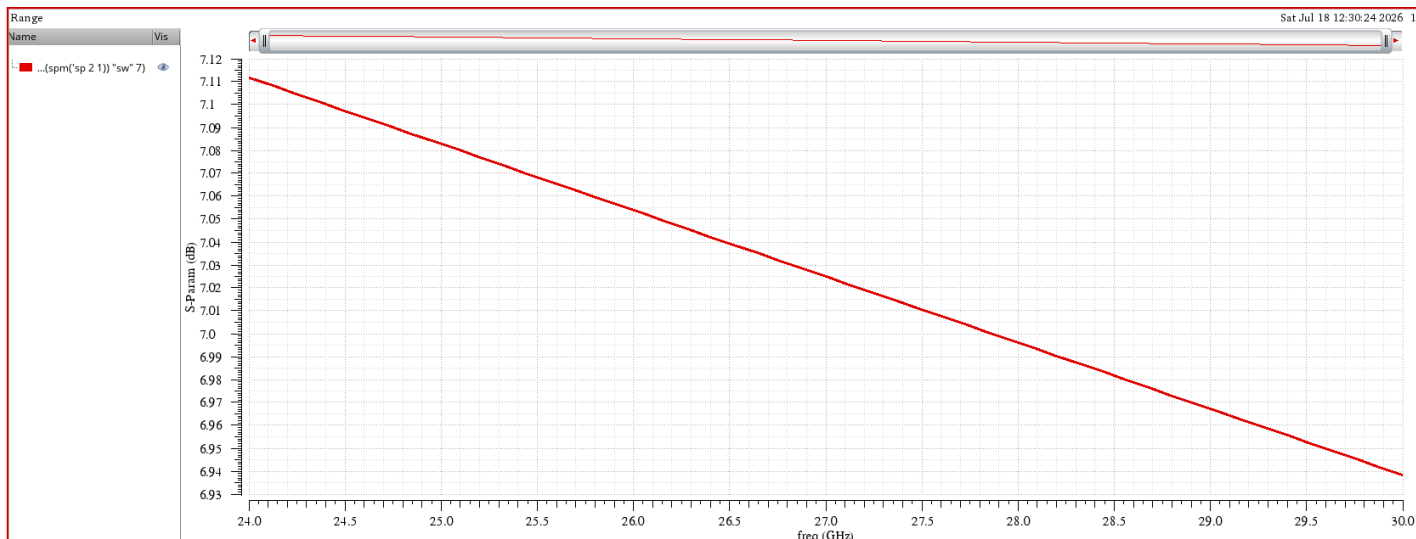


Fig. III-5: Attenuation range (S21 at sw=0 minus S21 at sw=7) vs. frequency, 24–30 GHz — ≈ 7 dB across the band, matching the gain-range spec.

8.4 Phase error and DNL @ 28 GHz

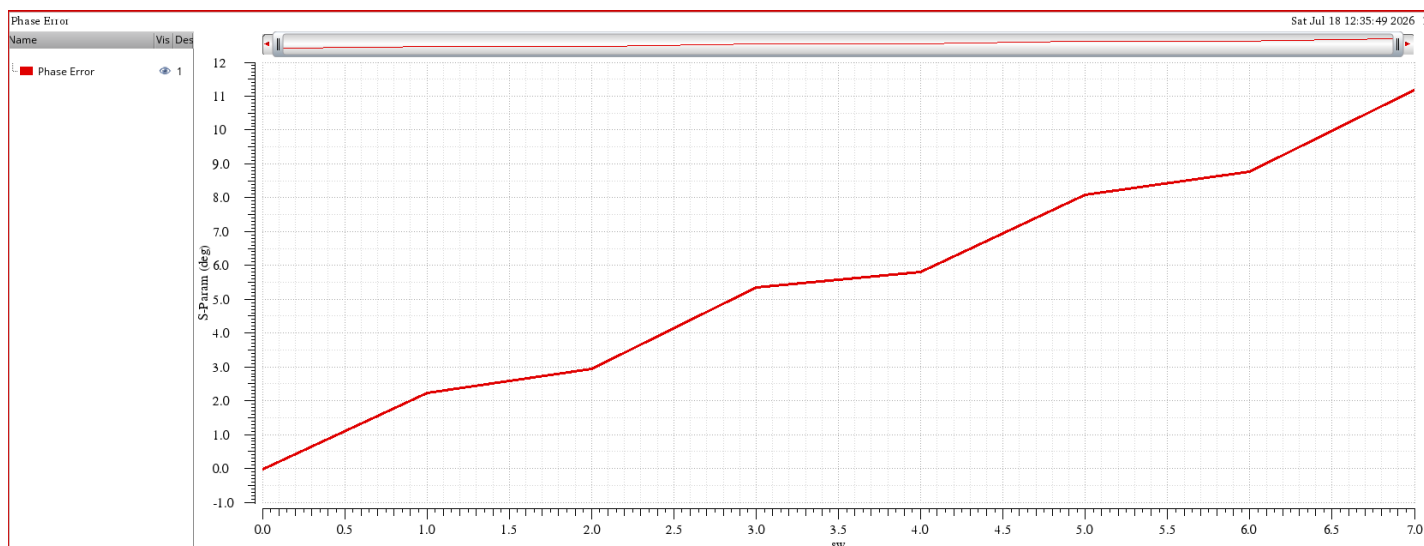


Fig. III-6: Phase error (relative to sw = 0) vs. attenuation state, at 28 GHz.

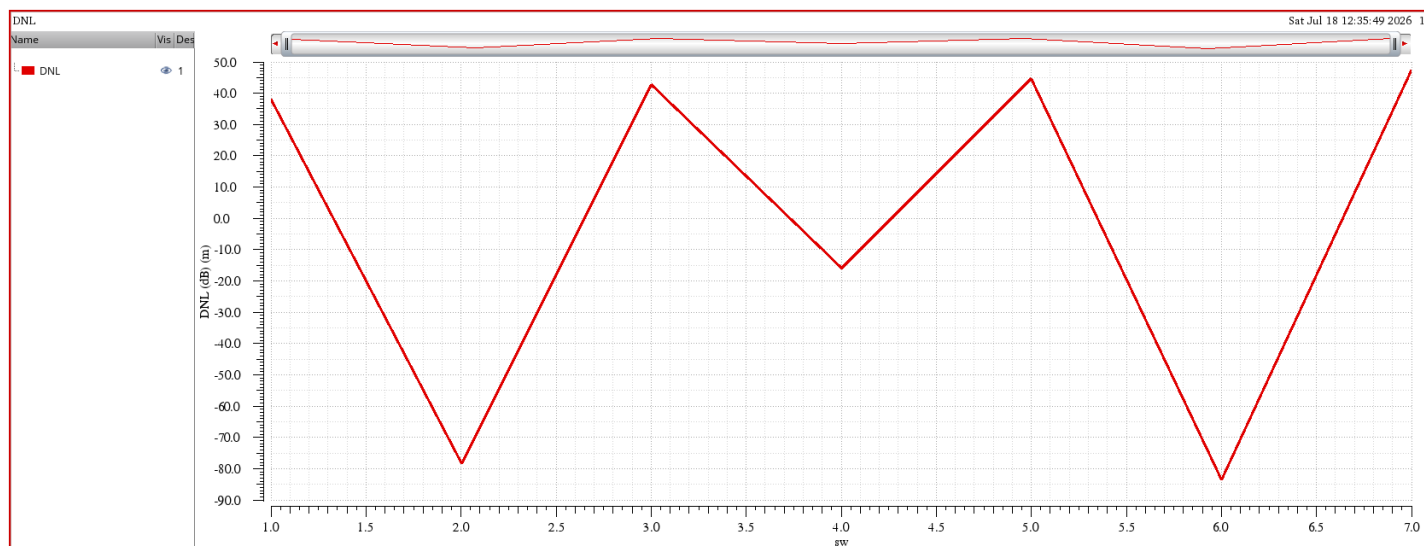


Fig. III-7: DNL vs. attenuation-state transition, at 28 GHz.

8.5 Linearity (OP1dB)

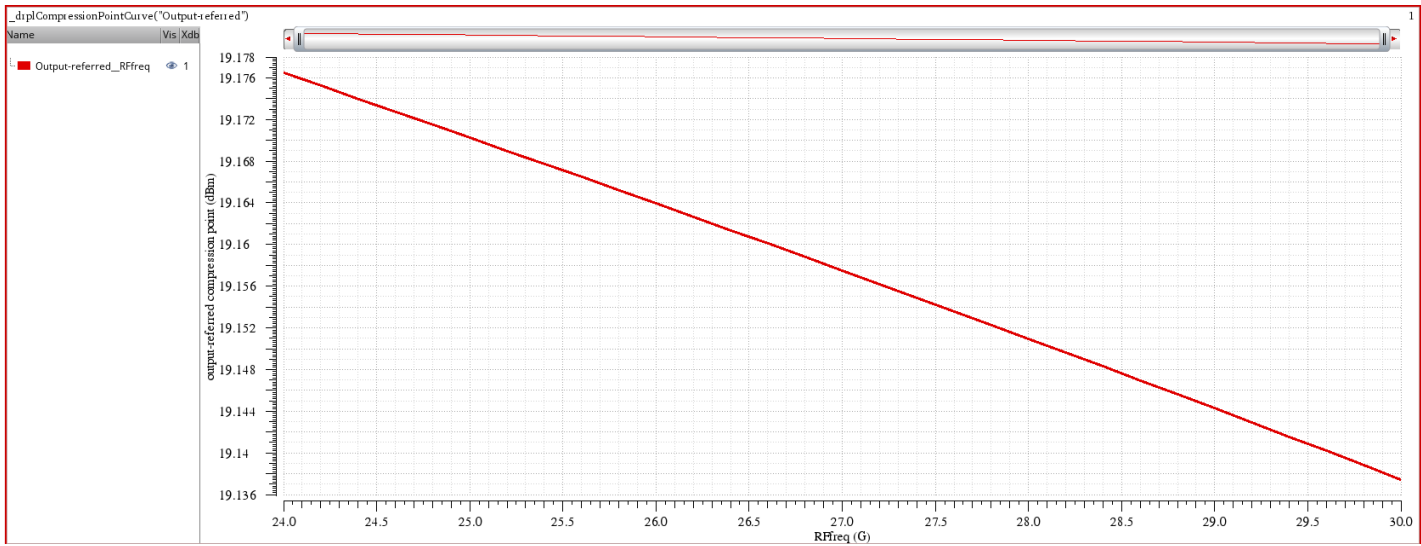


Fig. III-8: Output-referred P1dB (OP1dB) vs. frequency, 24–30 GHz, at the minimum-attenuation state ($sw = 0$).

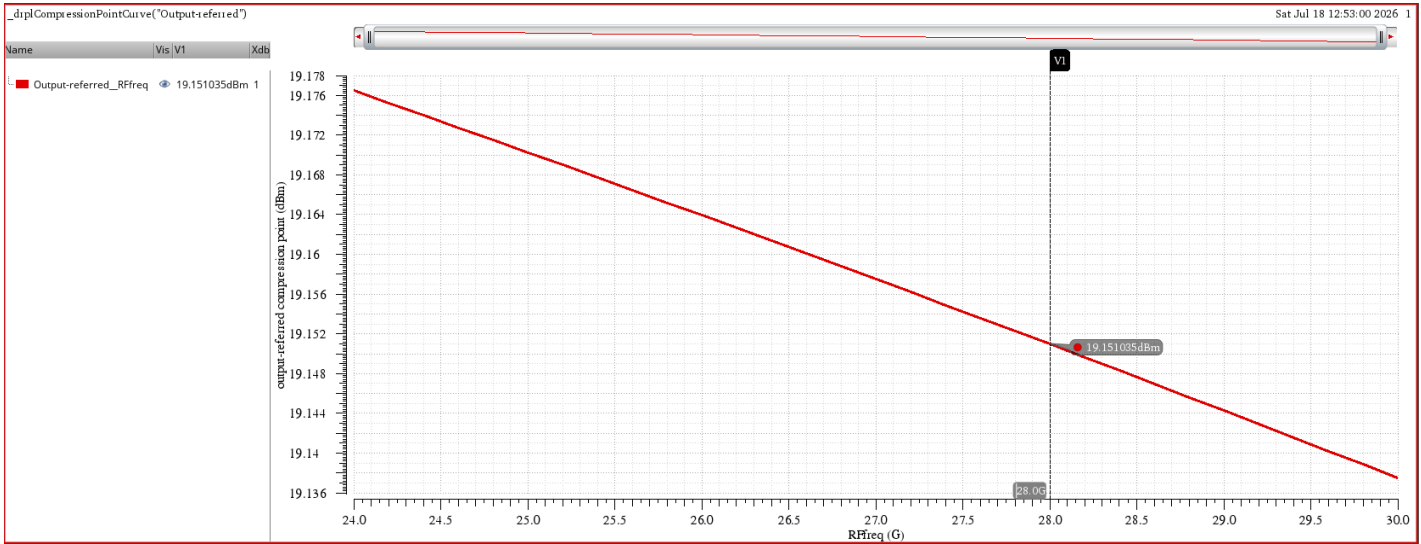


Fig. III-9: OP1dB marker read-out at 28 GHz, minimum-attenuation state ($sw = 0$) — ≈ 19.15 dBm.

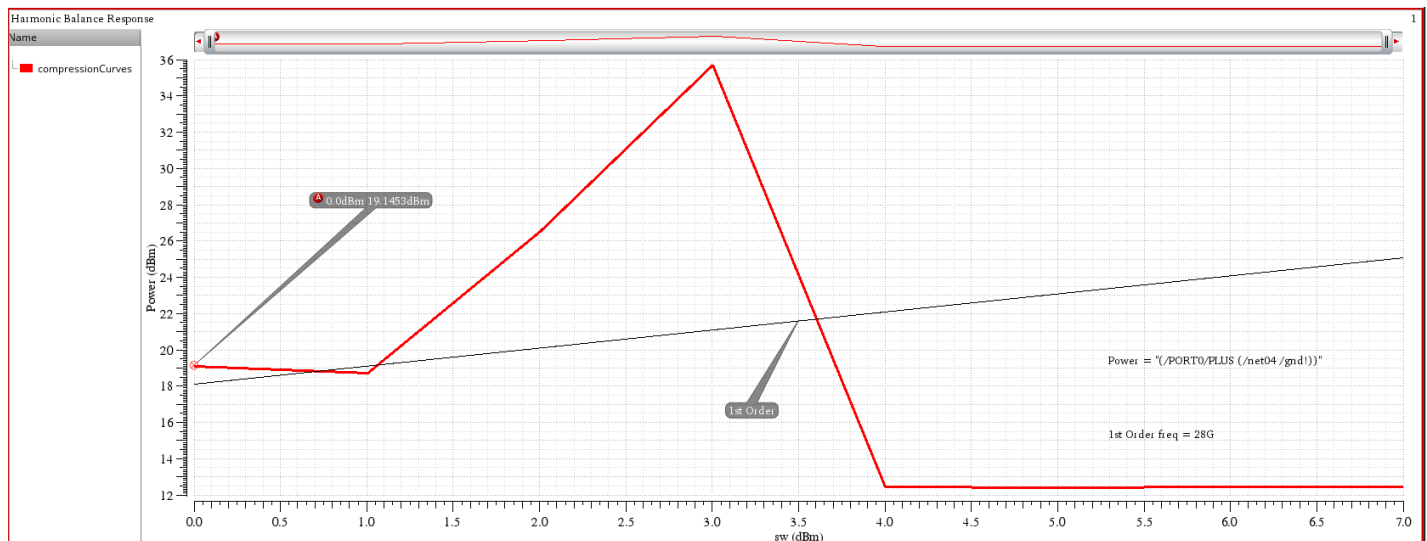


Fig. III-10: Harmonic-balance compression curves (OP1dB) across attenuation states, at 28 GHz.

8.6 DSA summary

The 3-bit (8-state) switched attenuator delivers the specified 7-dB range in 1-dB steps across 24–30 GHz with return loss and phase error within reasonable bounds, and OP1dB of roughly 19 dBm at the minimum-attenuation (through) state at 28 GHz. As expected for a passive resistive-switch attenuator, noise figure at a given state is close to that state's insertion loss, and linearity is high because the block is passive rather than gain-generating.

9. Overall Conclusion

The survey in Part I shows two published strategies for mmWave gain control in CMOS — noise/power-optimized active gain steering (Chang & Lin) and phase-invariant differential attenuation (Wu et al.). The author's own designs in Parts II and III sit closer to the second philosophy: both the VGA (Part II) and the DSA (Part III) use passive switched-resistor attenuation added to (or built from) an existing LNA/matching network, trading a modest, well-controlled gain/attenuation range and good phase/DNL/INL linearity for simplicity, rather than pursuing the more aggressive active current-steering or differential-cancellation techniques used in the surveyed papers. This is a reasonable choice for a first implementation, and the surveyed techniques — particularly Wu et al.'s differential Cgd-cancellation and isolation-enhanced layout — represent a natural next step if tighter phase invariance is needed at wider bandwidth in a future revision.